ECON 131 - INTERNATIONAL FINANCIAL MARKETS

Pavitra Kanagaraj

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KEY TAKEAWAYS: LECTURES 8, 9 & 10

LECTURE 8: DEBT FINANCING & MNC CURRENCY MODELS

- **Global Financing Mix**:
  - **U.S.**: Equity and bond markets dominate(market-oriented).
  - **Japan**: Bond market dominates.
  - **Europe**: Bank loans dominate (bank-oriented).
- **All-In-Cost (AIC)**: The internal rate equating future debt service to **net proceeds** (Price minus issuance fees):

$$\text{Net Proceeds} = \sum_{t=1}^{n} \frac{\text{Coupon}_t}{(1+\text{AIC})^t} + \frac{\text{Face Value}}{(1+\text{AIC})^n} \implies \text{AIC} > \text{YTM}$$

- **Debt Denomination Strategies for MNCs**:
  - **Centralized**: Parent issues home-currency debt; creates currency mismatch with foreign operating flows.
  - **Decentralized**: Subsidiaries borrow in local currencies; establishes a natural operational hedge.

LECTURE 9: BRADY BONDS & SOVEREIGN DEBT CRISES

- **Brady Bonds**: Dollar-denominated sovereign debt created to restructure defaulted bank loans in developing economies.
- **Collateral Structure**:
  - Principal (and often rolling coupons) collateralized by U.S. Treasury zero-coupon bonds.
  - Non-collateralized coupons reflect pure sovereign credit risk.
- **Stripped Price & Country Spread**:

$$P_{\text{stripped}} = P_{\text{bond}} - PV(\text{Collateral}) = \sum_{t=1}^{n} \frac{\text{Coupon}_t^{\text{risky}}}{(1 + y_{\text{stripped}})^t}$$

$$\text{Country Spread} = y_{\text{stripped}} - r_{\text{US}}$$

- **Default Probability (p)**: Implied from stripped cash flows via risk-neutral survival pricing: $(1-p)^t$.

LECTURE 10: EQUITY VALUATION & GORDON GROWTH

- **Valuation by Comparables**: Price-to-Earnings (P/E) and Market-to-Book (M/B) ratios.
- **Discounted Cash Flow (DCF)**: Intrinsic stock price equals the present value of future dividends and terminal share value:

$$P_0 = \sum_{t=1}^{n} \frac{D_t}{(1+r)^t} + \frac{P_n}{(1+r)^n}$$

- **Gordon Growth Model (Constant Growth g)**:

$$P_0 = \frac{D_1}{r-g}$$

- **Fundamental Growth Rate (g)**:

$$g = \text{Retention Ratio} \times \text{ROE} = (1 - \text{Payout Ratio}) \times \text{ROE}$$

LECTURE 10: CAPM & PORTFOLIO RISK/RETURN

- **Capital Asset Pricing Model (CAPM)**:

$$E(r_i) = r_f + \beta_i[E(r_m) - r_f], \quad \beta_i = \frac{\text{Cov}(r_i, r_m)}{\sigma_m^2}$$

- **Market Pricing Rule**:
  - $E(r) > r_{\text{CAPM}} \implies \alpha > 0 \implies$ **Underpriced** (attractive buy).
  - $E(r) < r_{\text{CAPM}} \implies \alpha < 0 \implies$ **Overpriced** (poor risk/return).
- **Two-Asset Portfolio Risk and Return**:

$$E(r_p) = w_A E(r_A) + w_B E(r_B)$$

$$\sigma_p = \sqrt{w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho_{AB}}$$

- **Sharpe Ratio**: Excess return per unit of total portfolio risk:

$$\text{Sharpe Ratio} = \frac{E(r_p) - r_f}{\sigma_p}$$

PS4

- **Problem 1 (Brady Bond)**:
  - Mexico Brady bond: $T = 7$ years, coupon = 11%, face value = \$100, price = 95.
  - Only par value is collateralized by U.S. Treasuries ($r_{US} = 4.0\%$ flat).
  - Tasks: (a) PV of collateral, (b) stripped YTM equation, (c) default probability p equation.
- **Problem 2 (CAPM)**:
  - $r_f = 2.0\%$, $E(r_m) = 8.0\%$.
  - Tasks: (a) Market risk premium, (b) required return for $\beta = 2.0$, (c) mispricing for $\beta = 0.6$ offering 7.0%.
- **Problem 3 (Sharpe Ratio)**:
  - $w_A = 0.30$, $w_B = 0.70$; $E(r_A) = 10\%$, $E(r_B) = 20\%$.
  - $\sigma_A = 0.60$, $\sigma_B = 0.40$, $\rho_{AB} = 0.10$, $r_f = 1.0\%$. Compute Sharpe ratio.

PARALLEL PROBLEM 1: BRADY BOND VALUATION

PARALLEL PROBLEM 1: SETUP & COLLATERAL PV

- **Parallel Scenario (Modified Parameters)**:
  - A Brady bond has $T = 5$ years remaining, face value = \$100, and market price $P = 92$.
  - Annual coupon = 10% (\$10 per year).
  - Only the \$100 principal at maturity is collateralized by U.S. Treasuries.
  - U.S. risk-free spot rate is $r_{US} = 5.0\%$ for all maturities.
- **(a) Present Value of Collateral (PV_{coll})**:

$$PV_{coll} = \frac{\text{Face Value}}{(1 + r_{US})^T} = \frac{\$100}{(1 + 0.05)^5} = \frac{\$100}{1.27628} = \$78.35$$

- **Stripped Price of Sovereign Risk**:

$$P_{stripped} = P_{bond} - PV_{coll} = \$92.00 - \$78.35 = \$13.65$$

PARALLEL PROBLEM 1: STRIPPED YTM & DEFAULT RISK

- **(b) Equation for Stripped Yield-to-Maturity (y_s):**
  - The stripped yield equates the stripped price to the present value of uncollateralized coupons:

$$P_{\text{stripped}} = \sum_{t=1}^{T} \frac{\text{Coupon}}{(1+y_s)^t} \implies 13.65 = \sum_{t=1}^{5} \frac{10}{(1+y_s)^t}$$

  - Solves to $y_s = 67.76\%$ (stripped country spread $= 67.76\% - 5.0\% = 62.76\%$).
- **(c) Equation for Annual Default Probability (p):**
  - Under risk-neutral survival pricing with zero recovery:

$$P_{\text{stripped}} = \sum_{t=1}^{5} \frac{(1-p)^t \times \text{Coupon}}{(1+r_{\text{US}})^t} \implies 13.65 = \sum_{t=1}^{5} \frac{10 \times (1-p)^t}{(1+0.05)^t}$$

  - Solves to an implied default probability of $p \approx 37.41\%$ per year.

PARALLEL PROBLEM 2: CAPM & VALUATION

PARALLEL PROBLEM 2: MARKET RISK PREMIUM & REQUIRED RETURN

- **Parallel Parameters**:
  - Treasury bill (risk-free) rate: $r_f = 3.0\%$.
  - Expected return on the market portfolio: $E(r_m) = 9.0\%$.
- **(a) Market Risk Premium (*MRP*)**:

$$MRP = E(r_m) - r_f = 9.0\% - 3.0\% = 6.0\%$$

- **(b) Required Return for an Asset with $\beta = 1.5$**:
  - Using the Security Market Line (SML):

$$E(r_i) = r_f + \beta_i \times [E(r_m) - r_f]$$

$$E(r_i) = 3.0\% + 1.5 \times (6.0\%) = 3.0\% + 9.0\% = 12.0\%$$

PARALLEL PROBLEM 2: MISPRICING & ALPHA

- **(c) Evaluating an Investment with $\beta = 0.5$ Offering $E(r) = 5.0\%$:**
  1. Compute the CAPM benchmark (required return k):

$$k = r_f + \beta \times MRP = 3.0\% + 0.5 \times (6.0\%) = 6.0\%$$

  2. Compare offered expected return against CAPM required return:

$$\text{Offered Return } (5.0\%) < \text{Required Return } (6.0\%)$$

  3. Jensen's Alpha: $\alpha = 5.0\% - 6.0\% = -1.0\%$.
- **Conclusion**:
  - The asset is **overpriced** in the market.
  - Investors are paying too high a price for its future cash flows, resulting in an expected return that falls below the compensation required for its systematic risk.

PARALLEL PROBLEM 3: PORTFOLIO SHARPE RATIO

PARALLEL PROBLEM 3: RETURN & VARIANCE SETUP

- **Parallel Parameters**:
  - Portfolio weights: $w_A = 0.40$, $w_B = 0.60$.
  - Expected returns: $E(r_A) = 8.0\%$, $E(r_B) = 16.0\%$.
  - Standard deviations: $\sigma_A = 0.50$, $\sigma_B = 0.30$.
  - Correlation: $\rho_{AB} = 0.20$; Risk-free rate: $r_f = 2.0\%$.
- **Step 1: Portfolio Expected Return** $E(r_p)$:

$$E(r_p) = w_A E(r_A) + w_B E(r_B) = (0.40 \times 8\%) + (0.60 \times 16\%)$$

$$E(r_p) = 3.2\% + 9.6\% = 12.80\% = 0.128$$

PARALLEL PROBLEM 3: PORTFOLIO RISK & SHARPE RATIO

- **Step 2: Portfolio Variance σ_p^2:**

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho_{AB}$$

$$\sigma_p^2 = (0.40)^2(0.50)^2 + (0.60)^2(0.30)^2 + 2(0.40)(0.60)(0.50)(0.30)(0.20)$$

$$= (0.16 \times 0.25) + (0.36 \times 0.09) + (0.48 \times 0.03)$$

$$= 0.0400 + 0.0324 + 0.0144 = 0.0868$$

- **Step 3: Portfolio Standard Deviation σ_p:**

$$\sigma_p = \sqrt{0.0868} = 0.2946 \text{ (or } 29.46\%)$$

- **Step 4: Sharpe Ratio**:

$$\text{Sharpe Ratio} = \frac{E(r_p) - r_f}{\sigma_p} = \frac{12.8\% - 2.0\%}{29.46\%} = \frac{0.108}{0.2946} = \mathbf{0.3666}$$

SUMMARY CHECKLIST FOR STUDENTS

- **Brady Bond Collateralization**:
  - Principal collateral is default-free: discount at r_{US}, never at sovereign yield.
  - Stripped price = $P_{\text{market}} - PV_{\text{collateral}}$.
- **CAPM Mispricing Rule**:
  - $E(r) > k \implies \alpha > 0 \implies$ **Underpriced** (bargain).
  - $E(r) < k \implies \alpha < 0 \implies$ **Overpriced** (too expensive).
- **Portfolio Risk**:
  - Don't forget the covariance interaction term: $2w_A w_B \sigma_A \sigma_B \rho_{AB}$.
  - Sharpe ratio measures excess return per unit of total risk (σ_p).